# Supplementary Material: Map Growth and Policy Routing in Budgeted Online RGB-D 3DGS

Anonymous ICCV submission

Paper ID *****

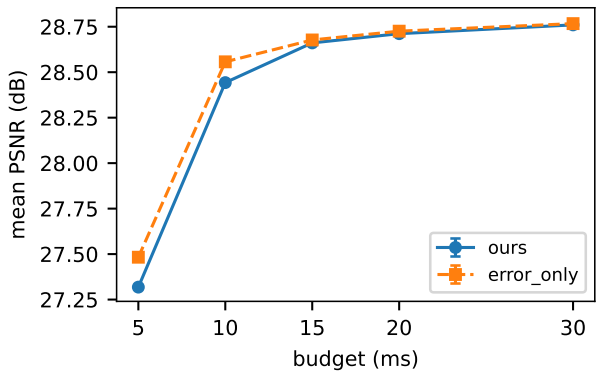



Figure S1. **Budget curve** (fr2_xyz, 30 frames×3, matched modeled budgets): policies are similar across the tested budget range at this short horizon. This experiment does not establish where the 150-frame policies separate as a function of budget.

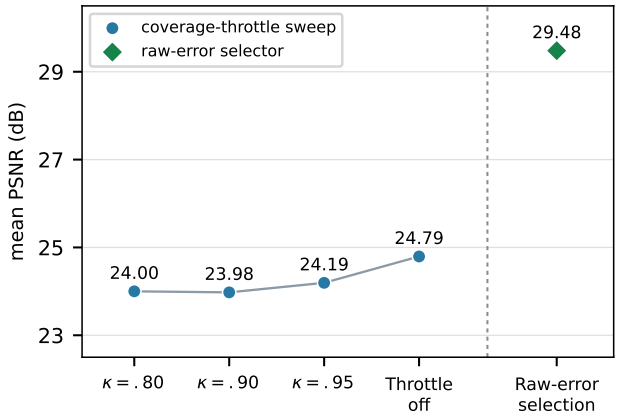



Figure S2. **Replica office0 ablation** (30 frames, $n = 3$). Disabling coverage throttling changes mean PSNR by 0.82 dB; switching from temporal×influence to raw-error changes it by a further 4.69 dB. These are descriptive short-run differences without a reported significance test.

## A. Short-horizon ablations

## B. Historical and Cross-System Context

The following tables use different substrates or protocols from the corrected matched-budget benchmark. **Historical component ablation (Tab. S1, pose-bugged substrate, 150 frames, 5 seeds).** Relative to A0, A2 (coverage) differs by +1.04 dB and A3 (coverage plus backlog) by +1.08 dB. The direct A3–A2 increment is +0.04 dB; no direct significance test is reported, and comparisons against A0 have Holm-adjusted $p = 0.25$. These numbers are not transferred to the corrected benchmark. A native local SplaTAM run on the same sequence (144f, seed 42: final 27.20 / mean 23.03 dB, SSIM 0.974, LPIPS 0.046, depth 1.26 cm, ATE 0.41 cm) uses a different resolution and tracking setup; these figures are contextual only and do not test compute dilution.

| Condition | Mean | Δ vs A0 | Map | Finding |
|---|---|---|---|---|
| A0 baseline | 16.07 | — | 140,998 | bloated |
| A1 + warmup | 15.32 | −0.75 | 140,038 | harmful |
| A2 + coverage | 17.11 | +1.04 | 44,865 | ns vs A0 |
| A3 + backlog | 17.15 | +1.08 | 43,824 | +0.04 vs A2 |
| A4 + learned | 14.53 | −1.54 | 38,317 | harmful |

Table S1. **Historical component ablation on the pose-bugged substrate** (5 seeds). Values and tests are not evidence for effects on the corrected benchmark. `results/step1_ablation_5seeds/`.

## C. Additional Evaluation

## D. Further Experiments

The following experiments would refine the mechanism and extend its scope. **Map-growth mediation:** vary continuous spawn rates and pruning strength while holding the candidate error threshold fixed; compare trajectories at matched

| Method | Replica | fr2_xyz | Protocol |
|---|---|---|---|
| MonoGS | 37.50 | 16.17 | full seq + refine |
| Point-SLAM | 35.17 | — | full seq + refine |
| RTG-SLAM | 32.79 | 17.08 | full seq + refine |
| SplaTAM | — | 24.50 | full seq + refine |
| SplaTAM (local run, 144f) | — | 27.20 final / 23.03 mean | SLAM 640×480, no refine |
| Photo-SLAM | — | 21.07 | full seq + refine |
| Ours (error_only) | 20–33 | 24.21 | budgeted, 100–150f |
| Ours (throttled) | 20–29 | **27.74** | budgeted, 100–150f |

Table S2. **Published-SOTA context (protocol-mismatched) and one local SplaTAM run.** Published results are shown for context, not as matched comparisons. The local run uses the same sequence but a different resolution and tracking setup; details are in the text.

| Method | fr2_xyz | fr1_xyz | Horizon | System? |
|---|---|---|---|---|
| Ours (GN VO) | 0.9 | 2.0 | 100f | no (baseline) |
| MonoGS | 1.71 | — | full | yes |
| SplaTAM | 0.50 | — | full | yes |
| Photo-SLAM | 0.63 | — | full | yes |

Table S3. **ATE RMSE (cm).** Rows 2–4 are published full-system numbers; ours is a decoupled odometry sanity check on fixed horizons.

| Policy | Mean | Final | SSIM | Map |
|---|---|---|---|---|
| Fixed OURS | 26.85 | 26.07 | 0.9244 | 78,825 |
| Routed (err) | 30.17 | 28.02 | 0.9584 | 115,964 |
| Δ | +3.32 ✓ | +1.95 ✓ | +0.034 | — |

Table S4. **Adaptive validation, Replica `office0`, 99 frames × 8 seeds.** Router picks `error_only` (holes ≈ 0.02): $p = 0.0078$, $d_z = 71.9$. No fixed policy wins both substrates.

map size and depth quality. A budget proportional to current map size would also test whether the fourfold rescue persists under a less coarse intervention. **Router generalization:** obtain a third sensor domain and scenes outside the reported set for threshold calibration; then test structured missing-depth artifacts and geometry-preserving TUM hole repair. The present random-hole and valid-depth-smoothing interventions did not reverse the preferred policy. **Longer mapping evaluation:** use sequences of at least 500 frames, multiple start windows, a real tracker, and depth or geometric metrics. **Corrected utility labels:** regenerate counterfactual labels on the corrected pose substrate before drawing conclusions about learned ranking. **Finer-grained adaptivity:** the 7-scene routing table shows motion/exploration regime matters within a sensor (fr1_room, fr1_rpy trail by $\approx -1.1$ dB); extending the observable cue beyond hole-rate (e.g. optical-flow magnitude, coverage velocity) could route rapid-motion frames to instantaneous error while keeping temporal smoothing elsewhere.

## E. Hole-rate policy routing (auxiliary)

Tables S4–S6 come from earlier code revisions, before the evaluator correction (standard windowed SSIM; alpha-normalized depth). Their SSIM and depth values are not comparable with the main-paper tables, and their PSNR may differ by a few tenths of a dB. Main-paper scene dependence motivates asking whether an observable cue can pick the better policy. We report this as an auxiliary analysis; it does not support a mechanism claim. **Hole-rate policy routing.** Raw depth hole-rate separates the evaluated TUM and Replica datasets: TUM 0.18–0.31 vs Replica $\approx 0.000$ (300–500×). Since the datasets also differ in rendering, pose quality, and scene content, this association does not establish that sensor noise causes the policy ranking to change. The score $n = \mathrm{clip}(2.5 \cdot \mathrm{holes} + 1.0 \cdot \mathrm{edge})$ routes $n \le 0.10 \rightarrow$ raw-error selection ($\kappa = 0.98$) and $n \ge 0.30 \rightarrow$ temporal×influence + throttling ($\kappa = 0.90$), computed pre-hygiene in milliseconds on CPU. Formally, with $h_t$ the raw missing-depth rate and $e_t$ the raw edge density of frame $t$:

$$n_t = \min(1,\ 2.5h_t + 1.0e_t), \quad \pi_t = \begin{cases} \texttt{err_only} & n_t \le 0.10 \\ \text{blend} & 0.10 < n_t < 0.30 \\ \texttt{temporal} & n_t \ge 0.30 \end{cases} \tag{1}$$

with coverage knee $\kappa \in \{0.98, 0.90\}$ tied to the chosen arm. The score and thresholds were chosen after inspecting five development scenes (Fig. S3), including `fr1_xyz`. That scene is therefore development evidence for the router, although its separate 150-frame mapping-policy comparison uses frozen settings. A seven-scene audit is reported separately in Tab. S6; scene-level independence must be tracked separately from seed-level repeats. Two earlier features failed held-out separation and are reported as such: post-hygiene invalid ratio (conflates far-but-valid geometry — 18% of Replica room0 exceeds 4 m — with holes) and Laplacian texture energy (measures sharpness, inverted: crisp synthetic scores *higher*). Operating on pre-hygiene raw depth was the fix (Fig. S3). A mixed-regime test injects 30% holes mid-trajectory on Replica and the router changes arms on 49/50 frames at the transition. This verifies feature-triggered switching in that test; end-to-end PSNR trails fixed `error_only` by 0.85 dB, so a quality benefit on mixed real sequences is not established.

**Policy choice differs across the two evaluated substrates (Tab. S4).** On synthetic Replica the fixed com-

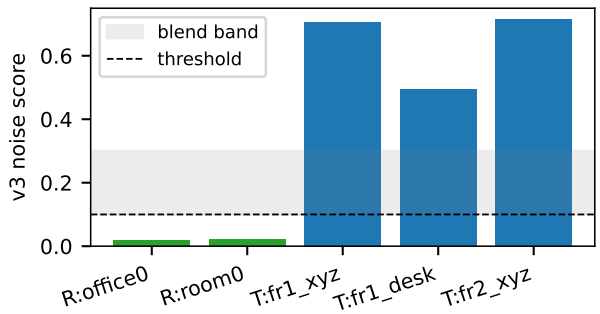



Figure S3. **Depth score on five development scenes.** The five plotted scenes include `fr1_xyz`; its mapping-policy comparison is distinct from independent router validation. The seven-scene router audit appears in Tab. S6. The separation shown here measures dataset association, not sensor-noise causality.

| Scene | Level | Error | Temporal | Δ |
|---|---|---|---|---|
| Replica office0 | 0% | 30.18 | 26.84 | −3.33 |
| | 15% | 30.32 | 26.75 | −3.58 |
| | 30% | 30.31 | 26.62 | −3.69 |
| TUM fr2_xyz | 0 | 24.12 | 27.69 | +3.57 |
| | 0.5 | 24.10 | 27.64 | +3.54 |
| | 1.0 | 24.21 | 27.62 | +3.41 |

Table S5. **Within-scene depth interventions** (150 frames on TUM, 99 on Replica, × 3 seeds). Δ is temporal×influence with coverage minus unthrottled raw error, measured on identical perturbed frames. Replica also receives Gaussian depth noise with standard deviation 0.05 times the hole fraction in meters. TUM smoothing acts only on valid depth after the standard hygiene stage; it does not remove remaining holes. No level reverses policy ordering. Full paired intervals: `results/depth_intervention/analysis.json`.

| Scene | Routed PSNR | Fixed reference | Route |
|---|---|---|---|
| fr2_xyz 150f | 27.66 | 27.64 (ours) | temporal |
| office0 99f | 30.36 | 30.17 (err) | error_only |
| office1 100f | 33.11 | 33.48 (err) | error_only |
| room1 100f | 24.17 | 24.29 (err) | error_only |
| fr3_static 100f | 25.95 | 25.96 (ours) | temporal |
| fr1_room 100f | 15.50 | 16.54 (err) | temporal |
| fr1_rpy 100f | 19.93 | 21.06 (err) | temporal |

Table S6. **7-scene router audit.** Routed PSNR uses $n = 3$; fixed values are unpaired references. Office0 uses $n = 3$ routed vs $n = 8$ fixed. Five scenes are held out; two overlap development.

bined policy loses 3.32 dB; routing to raw-error selection recovers 3.32 [3.29, 3.35], $p = 0.0078$. **Seven-scene routing (Tab. S6, exploratory $n = 3$, no scene labels):** routed PSNR is close to fixed-policy reference values on five scenes and about 1.1 dB lower on exploration-heavy `fr1_room` and rapid-rotation `fr1_rpy`. These are unpaired, exploratory comparisons; the table reports references rather than inferential deltas. Domain dependence is thus finer than real-vs-synthetic: motion and exploration regime matter within a sensor, and the current router does not capture them — reported as-is. **Depth intervention (Tab. S5).** On Replica `office0`, raising random missing-depth rate to 30% with accompanying 1.5 cm Gaussian noise makes the frozen score exceed its temporal-route threshold, yet raw-error selection remains ahead by 3.69 [3.62, 3.78] dB across three seeds. On TUM `fr2_xyz`, bilateral smoothing of valid depth after the standard hygiene stage changes the temporal-minus-error advantage only from +3.57 to +3.41 dB. These interventions do not support a causal claim that missing depth alone determines the better policy. The Replica masks are independent pixel noise rather than a full Kinect artifact model; the TUM intervention leaves missing pixels masked. Thresholds were not recalibrated on independent scenes because every available TUM and Replica scene appears in the reported scene set, and no third sensor domain is present locally.

## F. Counterfactual utility and negative results (auxiliary)

**Measurement instrument.** Per-frame scheduling uses $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ and $\Delta Q$ measured by exact full-scene counterfactual renders (active edits embedded in all $N$ splats, avoiding depth-interleaving gradient bias of affine-background approximations). The oracle $U_i^\star = \Delta Q_i^{\text{exact}} / \Delta C_i$ is our measurement instrument, never a runtime input. It reveals: (i) negative utility in 2.7–38.7% of interventions by stratum (pose-fixed regeneration), with only 2/75 negative interventions at depth discontinuities; (ii) interaction is protocol-dependent — corrected joint-group measurements give $R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$ (relatively additive), while a separate pairwise protocol shows stronger interaction; pointwise top-$K$ is therefore treated as a reference, not a bound, without a universality claim; (iii) offline TwoHeadMLP rank signal ($\rho \approx 0.20$, NDCG@20 0.46, measured on pre-pose-fix labels whose absolute values are polluted but whose within-set method comparison remains valid) reverses to −2.51 dB closed-loop. Conditional-oracle selection numbers are withheld pending a rerun under fork-correct semantics. Pose-fixed counterfactuals are negative in 29/75 flat, 16/75 texture, 14/75 edge and 2/75 depth-discontinuity interventions (candidate-level proportions from five seeds; no cluster-aware intervals). Uniform random selection from the same stratified pools has an expected negative-intervention rate of 20.3% (95% seed-cluster bootstrap CI 16.0–24.3%), so the stratum range alone does not establish ranking quality; the per-candidate labels needed to compare the selected top-error subset were not saved. **Negative results. Learned utility reverses closed-loop** (−2.51 dB in the historical one-seed factorial on the pre-fix

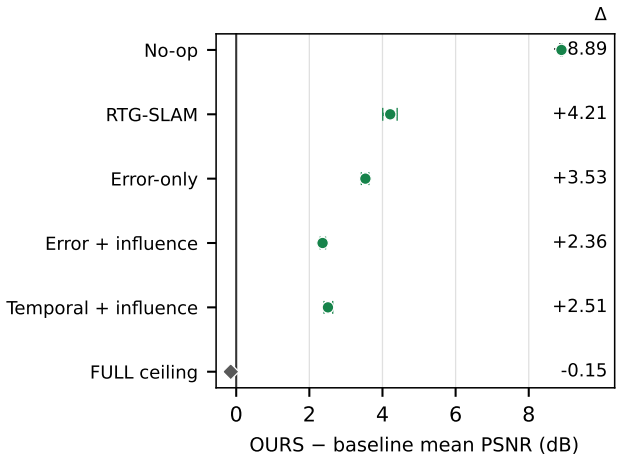



Figure S4. **Paired mean-PSNR differences** on fr2_xyz (150 frames, $n = 8$). Points show OURS − baseline; bars are 95% paired-bootstrap CIs. Positive favors OURS. FULL is 0.15 dB higher; all six contrasts pass Holm correction ($p = 0.0469$).

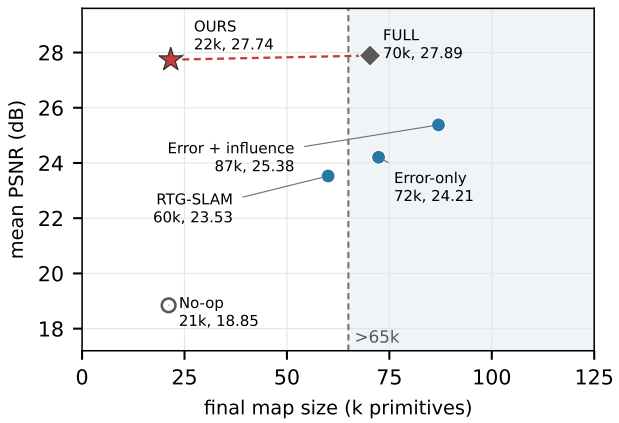



Figure S5. **Quality–map-size trade-off** on fr2_xyz (150 frames, $n = 8$). Among policies that optimize, only OURS and FULL lie on the frontier; FULL gains 0.15 dB with a 3.2× larger map. The four growing baselines hold 60–87k Gaussians; no-op stays compact at lower quality.

substrate; GradNorm beats the MLP pointwise $0.51 > 0.25$): offline labels ignore trajectory shift and inter-Gaussian coupling, so we ship no learned component. This result is separate from the corrected-pose factorial in Tab. 2 of the main paper. **Contextual re-ranking buys no online gain**: adaptive greedy shows no significant improvement over static pointwise selection in direct multi-seed budget sweeps (Wilcoxon n.s., CIs span zero) while incurring 424× selection overhead (76.36 vs 0.18 ms). Supporting rank-stability/oracle-gap numbers predate the fork-semantics fix and are withheld pending regen.

## G. Additional main-benchmark figures

## H. Per-scene breakdown of the combined policy (earlier code revision)

## I. Further diagnostics

**Dose-response (Tab. S8).** With free growth, PSNR rises monotonically with the budget multiplier on all four scenes (for example 24.24, 26.49, 27.82 dB on `fr2_xyz`), and the fraction of mature Gaussians updated per frame rises with it; their per-seed Spearman correlation over the three free-growth cells is +1.00 in every scene. Over all six cells it stays +0.92 and +0.75 on the TUM scenes, where a 23k cap is close to the natural map size, but not on Replica (+0.20 and −0.26), where the caps (60k, 40k) remove capacity and PSNR falls by 6–11 dB regardless of budget. Under a cap, ×2 and ×4 coincide once coverage reaches 1.00, because the budget exceeds what the capped map can use. Update coverage therefore tracks quality wherever capacity is not binding. These are within-scene interventions on raw-error selection. They show that the system is compute limited and that the limit tightens as the map grows, but more compute improves any compute-limited optimizer, so they do not by themselves show that growth causes the limitation; the cap cells address that only on TUM.

**Longer horizon (Tab. S9).** On the three TUM scenes whose sequences allow it (`fr2_xyz`, `fr1_xyz`, `fr1_desk`; 500 frames, two seeds), the unthrottled raw-error baseline degrades with horizon (window-mean PSNR 24.22 → 17.88, 21.50 → 19.32 and 22.49 → 16.95 dB from frames 1–150 to 301–500) and its final map reaches 116–136k Gaussians. The map-proportional budget (mean scale 5.4–6.7×) recovers +3.65 → +8.48, +3.03 → +5.77 and +1.42 → +5.92 dB over the same windows, whereas coverage throttling is not monotone (+3.47 → +4.85, +1.71 → +0.72, +0.47 → +1.17 dB). The gap to the unthrottled baseline therefore widens with horizon, consistent with dilution accumulating as the map grows, but with two seeds and three scenes this is descriptive, and the proportional budget spends several times more compute at 500 frames. Replica sequences are too short (99–119 frames) for this test.

**Held-out views (Tab. S10).** Every eighth frame is rendered but never mapped or optimized. The sign of every effect is unchanged on these frames but its size shrinks: the proportional budget's gain is +2.18, +1.69, +1.40 and +0.68 dB held out versus +2.85, +2.59, +2.58 and +1.44 dB on mapped frames (47–76% retained), and the train-minus-held-out gap grows with the budget (for example 3.29 → 4.47 dB on `office0`). Throttling's held-out changes are +2.13 and +1.04 dB on TUM and −0.27 and −0.57 dB on Replica. Part of the gain from more optimization is therefore specific

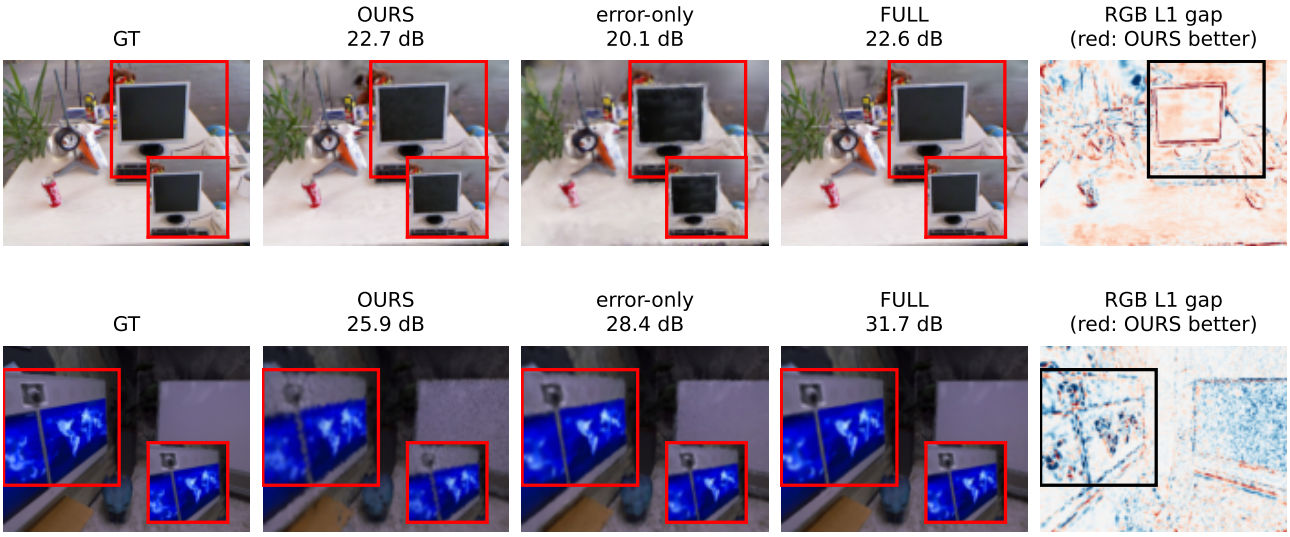



Figure S6. **Qualitative comparison across both evaluated substrates.** Top: an intermediate TUM `fr2_xyz` view (90), where OURS has lower per-view error than `error_only`; bottom: Replica `office0` view 98, where the ranking reverses. Columns show ground truth, OURS, `error_only`, FULL, and signed RGB $L_1$ error gap (red: OURS has lower error). Insets mark the same crop in each row. This figure comes from an earlier code revision. PSNR labels are per-view values from seed 42, shown only to illustrate the visual comparison; they are not sequence means.

| Scene | Frames | OURS | `error_only` | Δ | Map (k) | $n$ |
|---|---|---|---|---|---|---|
| fr2_xyz (TUM) | 150 | 27.74 | 24.21 | +3.53 ✓ | 22/ 72 | 8 |
| fr1_xyz (TUM) | 150 | 23.47 | 21.09 | +2.38 ✓ | 39 / 81 | 8 |
| fr1_desk (TUM) | 150 | 22.55 | 22.34 | +0.21 | 27 / 56 | 3 |
| fr1_room (TUM) | 100 | 15.53 | 16.54 | −1.01 | 92 / 97 | 3 |
| fr1_rpy (TUM) | 100 | 19.92 | 21.06 | −1.14 | 45 / 62 | 3 |
| fr3_static (TUM) | 100 | 25.96 | 25.88 | +0.08 | 13 / 18 | 3 |
| office0 (Rep.) | 99 | 26.85 | 30.17 | −3.32 ✓ | 79 / 116 | 8 |
| office1 (Rep.) | 100 | 29.25 | 33.46 | −4.21 ✓ | 42 / 74 | 8 |
| office2 (Rep.) | 99 | 22.08 | 24.67 | −2.59 | 70 / 117 | 3 |
| office3 (Rep.) | 99 | 22.45 | 24.47 | −2.02 | 77 / 150 | 3 |
| office4 (Rep.) | 99 | 23.58 | 25.93 | −2.36 | 81 / 141 | 3 |
| room0 (Rep.) | 99 | 20.85 | 22.78 | −1.93 | 82 / 150 | 3 |
| room1 (Rep.) | 100 | 20.42 | 24.29 | −3.87 | 47 / 59 | 3 |
| room2 (Rep.) | 99 | 22.86 | 25.88 | −3.03 | 78 / 146 | 3 |

Table S7. **Per-scene combined-policy breakdown, 14 sequences.** ✓ marks the reported significant comparisons at $n = 8$; other entries are exploratory. The combined policy wins the two confirmatory TUM comparisons but loses on both confirmatory Replica comparisons; remaining scenes have exploratory sample sizes.

to the mapped views. The held-out frames lie on the mapped trajectory, so this proxy does not test generalization to new trajectories.

**Estimated poses (Tab. S11).** We also map with poses from a frame-to-frame dense RGB-D Gauss-Newton tracker (ATE 1.24 and 2.45 cm; it does not use the map, so ATE is shared across conditions). PSNR at the tracked pose improves by +4.09 to +4.44 dB on `fr2_xyz` and +3.12 to +4.48 dB on `fr1_xyz`, more than with ground-truth poses. Rendering the same maps from the ground-truth pose removes almost all of this (+0.02 to +0.03 and +0.15 to +0.22 dB; baseline 21.88 and 17.58 dB versus 27.35 and 23.75 dB at the tracked pose), because pose error rather than optimization budget then limits the alignment. Gains in self-consistent quality therefore do not imply better alignment with the true camera. In a map-based SLAM system a better map could also improve tracking, which this map-independent tracker cannot show.

| Scene (cap) | Free map growth ×1 | Free map growth ×2 | Free map growth ×4 | Hard map cap ×1 | Hard map cap ×2 | Hard map cap ×4 | $\rho_{\text{all}}$ | $\rho_{\text{free}}$ |
|---|---|---|---|---|---|---|---|---|
| fr2_xyz (23k) | 24.24 (0.38) | 26.49 (0.58) | 27.82 (0.98) | 26.90 (0.73) | 27.85 (1.00) | 27.85 (1.00) | +0.92 | +1.00 |
| fr1_xyz (23k) | 21.53 (0.29) | 22.78 (0.55) | 24.45 (0.89) | 22.52 (0.65) | 23.63 (1.00) | 23.63 (1.00) | +0.75 | +1.00 |
| office0 (60k) | 30.61 (0.17) | 32.25 (0.34) | 33.43 (0.72) | 24.37 (0.24) | 25.25 (0.49) | 26.07 (1.00) | +0.20 | +1.00 |
| office1 (40k) | 33.85 (0.18) | 34.88 (0.33) | 35.42 (0.68) | 22.79 (0.38) | 22.85 (0.75) | 22.90 (1.00) | -0.26 | +1.00 |

Table S8. **Dose-response in optimization budget, 8 seeds.** Raw-error selection, no throttling; entries are mean PSNR (dB) with the mature-update coverage in parentheses, for a budget of $15 \times \{1, 2, 4\}$ ms with the spawn allowance fixed at 15 ms, with free growth or a hard Gaussian cap. $\rho$ is the per-seed Spearman correlation between PSNR and mature-update coverage over the cells (all six, or the three free-growth cells), averaged over seeds. Once coverage reaches 1.00 extra budget is unused and PSNR saturates. On Replica the caps remove needed capacity (PSNR 6–11 dB lower at equal budget), so the cross-regime correlation does not hold there.

| Scene | Frames | Base | Cov. | Prop. |
|---|---|---|---|---|
| fr2_xyz | 1–150 | 24.22 | +3.47 | +3.65 |
| | 151–300 | 18.70 | +6.07 | +9.30 |
| | 301–500 | 17.88 | +4.85 | +8.48 |
| fr1_xyz | 1–150 | 21.50 | +1.71 | +3.03 |
| | 151–300 | 19.74 | +1.88 | +5.37 |
| | 301–500 | 19.32 | +0.72 | +5.77 |
| fr1_desk | 1–150 | 22.49 | +0.47 | +1.42 |
| | 151–300 | 18.98 | +2.52 | +5.05 |
| | 301–500 | 16.95 | +1.17 | +5.92 |

Table S9. **Long horizon (500 frames, two seeds).** Mean PSNR of the unthrottled raw-error baseline (Base) and the change from coverage throttling (Cov.) and the map-proportional budget (Prop., mean scale 5.4–6.7×) in three frame windows. The baseline degrades with horizon and the gains grow; coverage throttling is not monotone. Replica sequences are too short for this test.

| Mapped / held-out | fr2_xyz | fr1_xyz | office0 | office1 |
|---|---|---|---|---|
| Coverage throttling | +2.73/+2.13 | +1.50/+1.04 | −0.62/−0.27 | −0.93/−0.57 |
| Spawn cap ×0.2 | +2.68/+2.11 | +1.38/+0.96 | −0.77/−0.50 | −1.08/−0.54 |
| Budget ∝ map | +2.85/+2.18 | +2.59/+1.69 | +2.58/+1.40 | +1.44/+0.68 |
| Gap, unthrottled | 0.87 | 0.98 | 3.29 | 2.79 |
| Gap, budget ∝ map | 1.53 | 1.87 | 4.47 | 3.55 |

Table S10. **Mapped versus held-out views, change in mean PSNR (dB) relative to unthrottled raw-error selection, four seeds.** Every eighth frame is rendered but never mapped (a same-trajectory proxy, not an independent trajectory). The last rows give the train-minus-held-out PSNR gap.

| Scene | Condition | Tracked pose | GT pose |
|---|---|---|---|
| fr2_xyz (ATE 1.24 cm) | unthrottled | 27.35 | 21.88 |
| | coverage | +4.31 | +0.02 |
| | spawn cap ×0.2 | +4.09 | +0.03 |
| | budget ∝ map | +4.44 | +0.02 |
| fr1_xyz (ATE 2.45 cm) | unthrottled | 23.75 | 17.58 |
| | coverage | +3.40 | +0.15 |
| | spawn cap ×0.2 | +3.12 | +0.15 |
| | budget ∝ map | +4.48 | +0.22 |

Table S11. **Mapping with estimated poses, three seeds.** PSNR (dB) of the unthrottled baseline and the change for each condition, evaluated at the tracker's pose and when the same map renders from the ground-truth pose. The tracker is a frame-to-frame RGB-D Gauss-Newton odometer that does not use the map.

**Closed-loop map-based tracking (Tab. S12).** To let map quality feed back into tracking, each frame's pose is estimated by rendering the current map and optimizing a 6-DoF correction coarse to fine (render scales $1/4, 1/2, 1$ with $20/15/10$ Adam steps, learning rate $10^{-3}$, constant-velocity initialization) on a photometric plus depth L1 loss, as in SplaTAM-style tracking but without keyframes, loop closure or global refinement; the controllers then map with these poses. The settings were explored on the first 60 frames of the development scenes fr1_desk and fr3_structure_texture_near, not on test scenes; ATE there was 3.6–6.5 cm across the handful of settings tried (single runs, noisy), and we kept the cheaper $20/15/10$ schedule at 4.4 cm rather than the best, 3.6 cm. The tracker is weaker than the frame-to-frame odometer (1.9–2.4 cm on those frames), and tracking costs 130–170 ms per frame, reported separately from mapping time and not subject to the TT timer. On fr1_room tracking fails (baseline ATE 0.97 m), so it is shown but excluded from the summary; that threshold (50 cm) was set after seeing the results. On the other four TUM test scenes, TT+VGG raises PSNR at the tracked pose by +0.10 to +2.83 dB (+1.56 dB mean), the same sign as with ground-truth poses, while ATE changes by between −1.61 and +1.50 cm (median −0.63 cm; lower in three of four) on a 6–14 cm baseline, which three seeds cannot separate from zero. The controllers therefore improve map fit at the tracked poses without a reliable effect on this tracker's accuracy. The local Replica sequences are too sparsely sampled for any tracker to work (median step 13–19 cm and 7–11° per frame, versus 0.3–1.1 cm and 0.3–0.7° on TUM; the frame-to-frame odometer gives ATE 1.4–4.7 m), so closed-loop results are TUM only.

**Value-gated growth versus a map-size budget (Tab. S13).** Taming 3DGS and Mini-Splatting control reconstruction through a Gaussian-count budget. We test that

| Scene | Base ATE | TT+VGG ΔPSNR | TT+VGG ΔATE |
|---|---|---|---|
| fr2_xyz | 7.5 cm | +2.83 | −1.61 cm |
| fr1_xyz | 6.4 cm | +1.89 | −0.92 cm |
| fr1_rpy | 12.6 cm | +1.42 | +1.50 cm |
| fr3_sitting_static | 14.0 cm | +0.10 | −0.34 cm |
| fr1_room† | 96.7 cm | −0.10 | +4.69 cm |

Table S12. **Closed-loop mapping with map-based tracking, three seeds.** Poses come from aligning each frame to the current map (Supp. text); the same controllers then map with the estimated poses. ΔPSNR (dB) is measured at the tracked pose, ΔATE is TT+VGG minus the unthrottled baseline. †Tracking diverged for every condition and the scene is excluded from the summary: over the four converged scenes TT+VGG changes PSNR by +1.56 dB on average and ATE by a median of −0.63 cm (largest |Δ| 1.61 cm).

| Scene | TT+VGG | TT + count cap | Count cap |
|---|---|---|---|
| replica_office0 | +1.31 | +0.83 | −0.01 |
| replica_office1 | +0.72 | −6.06 | −6.35 |
| replica_office3 | +1.44 | +0.42 | −0.32 |
| replica_office4 | +1.41 | +0.33 | −0.25 |
| replica_room0 | +0.99 | +0.32 | −0.23 |
| replica_room2 | +1.35 | +0.69 | −0.03 |
| tum_fr1_room | +0.82 | −3.70 | −3.69 |
| tum_fr1_rpy | +1.01 | −0.48 | −0.99 |
| tum_fr1_xyz | +1.12 | +0.27 | −0.08 |
| tum_fr2_xyz | +3.83 | +3.78 | +1.68 |
| tum_fr3_sitting_static | +0.06 | +0.81 | +0.72 |
| mean | +1.28 | −0.25 | −0.87 |

Table S13. **Value-gated growth versus a hard map-size budget of the same size.** PSNR change (dB) against the unthrottled baseline, two seeds per scene. The cap on the number of Gaussians equals TT+VGG's final map size on that scene and seed, so the cap is informed by the answer; "TT + count cap" also uses the time-targeted budget.

control inside our pipeline: a hard cap on the number of Gaussians equal to TT+VGG's final map size on the same scene and seed (so the cap uses information the controller must discover online), with and without the time-targeted budget, two seeds on the 11 test scenes. With the same time-targeted budget the cap averages $-0.25$ dB against the baseline, against $+1.28$ dB for TT+VGG, which is better on 10 of 11 scenes (median difference $+0.85$ dB). The cap fails because it stops growth when the map is full, wherever the new Gaussians would have been needed, whereas VGG decides per frame whether recent spawns were worth keeping; with the time-targeted budget it ends 6.1 dB below the baseline on office1 and 3.7 dB below on fr1_room. These are budget-matched controls, not the published methods, whose scoring and pruning rules we do not reimplement.

| Scene | Base ATE | TT+VGG ΔPSNR | TT+VGG ΔATE |
|---|---|---|---|
| office0 | | +3.69 | |
| office1 | | +3.37 | |
| office3 | | +2.76 | |
| office4 | | +0.14 | |
| room0 | | +1.39 | |
| room2 | | | |
| fr1_room | | | |
| fr1_rpy | | | |
| fr1_xyz | | | |
| fr2_xyz | | | |
| fr3_sitting_static | | | |

| Scene | $n$ | Base | Cap | TT | TT+cap | TT+VGG |
|---|---|---|---|---|---|---|
| office0 | 4 | 30.61 (86) | −0.72 (79) | +1.37 (99) | +0.79 (98) | +1.32 (99) |
| office1 | 4 | 33.87 (85) | −1.09 (80) | +0.84 (99) | −0.43 (99) | +0.77 (99) |
| office3 | 3 | 24.40 (88) | −0.46 (77) | +1.41 (100) | +0.86 (96) | +1.42 (100) |
| office4 | 3 | 25.44 (85) | −0.30 (76) | +1.42 (99) | +0.90 (95) | +1.42 (99) |
| room0 | 3 | 22.62 (87) | −0.38 (78) | +1.01 (99) | +0.75 (99) | +0.99 (99) |
| room2 | 3 | 25.55 (87) | −1.67 (77) | +1.40 (99) | −0.55 (99) | +1.33 (99) |
| fr1_room | 3 | 16.71 (99) | −0.17 (98) | +0.83 (101) | +1.01 (99) | +0.82 (102) |
| fr1_rpy | 3 | 21.10 (95) | +0.09 (93) | +1.07 (96) | +1.16 (84) | +1.02 (97) |
| fr1_xyz | 4 | 21.52 (101) | +1.59 (93) | +1.11 (94) | +2.90 (73) | +1.14 (95) |
| fr2_xyz | 4 | 24.06 (90) | +3.45 (81) | +2.94 (84) | +3.85 (60) | +3.88 (72) |
| fr3_sitting_static | 3 | 26.40 (69) | +0.11 (55) | +0.06 (56) | +0.09 (55) | +0.05 (57) |

Table S14. **Per-scene results, 11 held-out scenes.** Base: unthrottled PSNR (ms); other columns: change in PSNR (dB) with the mean frame time in ms in parentheses.

| Scene | $n$ | Base | Cap | TT | TT+cap | TT+VGG |
|---|---|---|---|---|---|---|
| fr1_desk | 3 | 23.09 (92) | −0.18 (90) | +1.02 (82) | +0.47 (65) | +1.04 (82) |
| fr1_room | 3 | 16.81 (97) | −0.11 (94) | +0.56 (93) | +0.42 (84) | +0.58 (94) |
| fr1_xyz | 3 | 22.88 (98) | +0.99 (90) | +0.74 (96) | +1.18 (69) | +0.87 (96) |
| fr2_xyz | 3 | 24.63 (90) | +1.58 (77) | +1.19 (87) | +1.80 (62) | +1.63 (85) |
| fr3_sitting_static | 3 | 28.35 (63) | +0.50 (50) | +0.34 (55) | +0.47 (51) | +0.41 (56) |

Table S15. **Per-scene results, five fresh frame windows.** Base: unthrottled PSNR (ms); other columns: change in PSNR (dB) with the mean frame time in ms in parentheses.

| Scene | Baseline $\times 1$/ $\times 2$/ $\times 4$ | $T = 70$ | $T = 85$ | $T = 100$ | $T = 12$ |
|---|---|---|---|---|---|
| fr2_xyz | 24.1@88 / 26.5@96 / 27.8@96 | 27.30@77 | 27.81@75 | 27.94@72 | 27.95@ |
| fr1_xyz | 21.5@100 / 22.8@108 / 24.4@119 | 21.76@92 | 21.86@92 | 22.67@95 | 23.93@1 |
| office0 | 30.6@86 / 32.3@105 / 33.4@135 | 30.64@83 | 30.79@86 | 31.93@99 | 33.00@1 |
| office1 | 33.8@86 / 34.9@107 / 35.4@138 | 33.87@84 | 33.96@86 | 34.64@99 | 35.24@1 |

Table S16. **TT+VGG at five time targets versus fixed budgets.** Mean PSNR (dB) @ mean end-to-end frame time (ms), four seeds. The baseline column gives free growth at modeled budgets $\times 1$, $\times 2$, $\times 4$.

## J. Controller details